Certified Feature Selection for Wasserstein Robust Hypothesis Testing

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Abstract

We study budgeted feature selection for binary hypothesis testing under class-conditional Wasserstein uncertainty. For finite, nonnegative, nondecreasing convex losses, an exact finite formulation constructs a whole-space test from every feasible solution. Scaling coordinate transport prices by the risk gives dual certificate fractions that are exact on binary selections. Every chosen subset receives a full continuous solve, producing an actual feasible-test upper bound and a globally valid tangent. A cheap Gaussian-process predictor uses completed solves to rank a small binary pool; a global integer master supplies the certified lower bound. Predictions determine which subset to try, but never replace an inner solve or provide certificates. Verified solve accuracies imply finite termination at the requested risk precision, with an evaluation bound determined by the data movement weights. Population coverage requires separate calibration.

1 The selection problem

We can observe at most d of D features. The choice of features and the test using them must be optimized together: a good subset is one on which a test can keep both classwise losses small when the distributions move within their specified uncertainty sets.

Pool the observations from the two classes into $x_1, \dots, x_N \in \mathbb{R}^D$. Observation i has class $c_i \in \{0,1\}$ and weight $q_i > 0$. We encode its class by the sign $y_i = 2c_i - 1$: $y_i = -1$ for class zero and $y_i = +1$ for class one. Both c_i and y_i are fixed, observed labels, not optimization variables. Write $I_k = \{i : c_i = k\}$. The empirical distributions and uncertainty sets are

$$\widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \sum_{i \in I_k} q_i = 1, \qquad \mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \quad k = 0, 1.$$

Here δ_x denotes unit mass at x. The distance W_1 is the minimum expected L^1 distance needed to transport one distribution to the other; the distributions have finite first moments. The radii $\theta_k > 0$ and feature scales are fixed before selection. Each class has total weight one, so the pooled weights sum to two.

For an integer budget $0 \le d \le D$, a binary vector z selects $S(z) = \{j : z_j = 1\}$, with $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. A measurable score g depends only on the selected coordinates. A positive score favors class zero; the losses are therefore $\Phi(-g)$ for class zero and $\Phi(g)$ for class one. The problem is

$$\begin{aligned} V(z) &= \inf_{g = g(x_{S(z)})} \left\{ \sup_{P \in \mathcal{U}_0} \mathbb{E}_P \Phi(-g) + \sup_{P \in \mathcal{U}_1} \mathbb{E}_P \Phi(g) \right\}, \\ \mathrm{OPT} &= \min_{z \in \mathcal{Z}_d} V(z). \end{aligned} \tag{1}$$

Thus $V(z)$ already includes the best possible test for the chosen features; selection does not hold a previously fitted test fixed.

The function $\Phi : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing. Nondecreasing means that a larger score against the true class cannot reduce its loss. We consider nonconstant losses with $\Phi(0) > 0$. If $\Phi(0) = 0$ or the loss is constant, the constant test already solves the problem. Specific formulas for Φ are not needed for the construction below; examples are given in Section 6. The constant score $g = 0$ provides the common benchmark

$$0 \le V(z) \le B := 2\Phi(0).$$

2 A finite problem that still designs the whole test

The same pooled observations serve as candidate target locations. To compare source i with target ℓ, use one vector of coordinatewise distances:

$$\mathbf{d}_{i\ell} = |x_i - x_\ell| \in \mathbb{R}^D_+, \qquad d_z(x_i, x_\ell) = \mathbf{d}_{i\ell}^\top z.$$

Absolute value is taken coordinatewise. The vector stores the geometry once; its inner product with z is the distance using the selected features.

Let u_ℓ be the score assigned to target ℓ. Let a_i be a loss allowance at source i, and let $\lambda_k \geq 0$ be the price per unit of transport for class k. Write $q = (q_i)$, $\theta = (\theta_0, \theta_1)^\top$, $\lambda = (\lambda_0, \lambda_1)^\top$, and $J(a, \lambda) = q^\top a + \theta^\top \lambda$.

Theorem 1 (Exact finite formulation)**.** *For every binary z,*

$$\begin{aligned} V(z) = \min_{a,u,\lambda} \quad & J(a,\lambda) \\ \textit{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z, \qquad i, \ell = 1, \dots, N, \\ & \lambda \geq 0, \qquad a, u \in \mathbb{R}^N. \end{aligned} \tag{2}$$

The minimum is attained. Every feasible solution yields a measurable test on the entire observation space, using only the selected coordinates, with robust risk at most $J(a, \lambda)$.

The inequality in (2) has a direct meaning: after moving from source i to target ℓ, the loss is at most its source allowance plus the transport price. The source retains its class c_i and sign y_i, while u_ℓ is the score at the target. Explicitly,

$$\Phi(y_i u_\ell) = \begin{cases} \Phi(-u_\ell), & c_i = 0, \\ \Phi(u_\ell), & c_i = 1. \end{cases}$$

Thus the loss uses the source class even when the target observation originally belongs to the other class; either class can send mass to any target location.

The whole-space conclusion is essential. It allows a finite feasible objective to be used as an upper bound for the original test problem. Its proof blends two source scores, one from each class. Convexity controls the blended losses, and the triangle inequality controls the required transport allowances. The construction and both directions of the equality are given in Appendix A; no particular inverse or closed form of Φ is used.

3 Use the risk to control which coordinates are available

Taking $\ell = i$ in (2) gives $a_i \geq \Phi(y_i u_i) \geq 0$. Consequently,

$$\theta^\top \lambda \leq J(a, \lambda). \tag{3}$$

The transport-price contribution is part of the total risk, so it cannot exceed that risk. This inequality supplies the bound needed to connect feature selection with continuous test design.

Introduce two vectors $w_0, w_1 \in \mathbb{R}^D_+$ of coordinate prices. On a binary selection we can take $w_{kj} = \lambda_k z_j$: an unobserved coordinate has price zero. To obtain a useful continuous extension, use the risk bound η to constrain these prices:

$$\begin{aligned} v(z) = \min_{\eta,a,u,\lambda,w_0,w_1} \quad & \eta \\ \text{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i}, \qquad \forall i, \ell, \\ & J(a,\lambda) \leq \eta, \qquad \lambda \geq 0, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0, 1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq \eta z. \end{aligned} \tag{4}$$

Vector inequalities hold coordinatewise; a, u, η are real variables. The first line designs the test using the coordinate prices. The next lines bound its risk and keep each coordinate price below the class price. The last line makes an unselected coordinate unavailable and ties the allowed prices to the risk being minimized.

For binary z, an optimizer of (2) enters (4) with $\eta = J$ and $w_k = \lambda_k z$, by (3). Conversely, a feasible point of (4) has zero prices on unselected coordinates and $w_k \le \lambda_k z$. The same a, u, λ therefore satisfy (2), with objective at most η. Hence $v(z) = V(z)$ at every binary selection. For fractional $z \in [0,1]^D$, (4) defines the extension used to construct lower bounds.

Fixing z makes (4) a convex problem. This alone does not prove that its value $v(z)$ is convex in z, since the last line contains the product ηz. The next step establishes that property and supplies information that remains useful when the feature subset changes.

4 One solved subset gives lower bounds for all subsets

Consider a nonnegative matrix Π with $\Pi_{i\ell}$ units of mass sent from source i to target ℓ. Its rows retain the source weights, $\Pi\mathbf{1} = q$. The masses received from the two classes and their coordinatewise movements are

$$p_{k\ell} = \sum_{i \in I_k} \Pi_{i\ell}, \qquad A_k(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} \mathbf{d}_{i\ell} \in \mathbb{R}^D_+.$$

At a target receiving both classes, one score must serve both. The least weighted loss that this common score can achieve is

$$\Psi(p_0, p_1) = \inf_s \{p_0 \Phi(-s) + p_1 \Phi(s)\}, \qquad H(\Pi) = \sum_\ell \Psi(p_{0\ell}, p_{1\ell}). \tag{5}$$

The infimum permits limiting scores when one class has zero mass. H measures the loss left after the best score is chosen at every target. It is nonnegative. Scaling all masses by $t \ge 0$ scales H by t; this property concerns the masses and does not require Φ itself to be homogeneous.

To use a matrix that may move more mass than the original budget allows, introduce a nonnegative correction vector $\mu \in \mathbb{R}^D_+$. The pair (Π, μ) must satisfy

$$\Pi \ge 0, \qquad \Pi\mathbf{1} = q, \qquad \mu \ge 0, \qquad \mathbf{1}^\top \big(A_k(\Pi) - \theta_k \mu\big)_+ \le \theta_k, \quad k = 0, 1. \tag{6}$$

Here $t_+ = \max(t, 0)$, applied coordinatewise to a vector. After $\theta_k \mu_j$ is deducted from movement in coordinate j, the remaining positive movements must fit inside the class budget θ_k. These conditions contain no z: the same pair can be used at any selection. The matrix need not be a transport plan inside the original uncertainty balls. The correction is what makes it useful as a risk lower bound.

The reason for the correction in the denominator can be seen directly. For $0 \le z \le 1$, (6) implies $z^\top A_k(\Pi) \le \theta_k(1 + \mu^\top z)$. For any feasible point of (2) on a binary selection,

$$\begin{aligned} H(\Pi) &\le \sum_{i,\ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ &\le q^\top a + \sum_k \lambda_k z^\top A_k(\Pi) \\ &\le J(a, \lambda) + (\mu^\top z) \theta^\top \lambda \\ &\le (1 + \mu^\top z) J(a, \lambda). \end{aligned}$$

The four steps use, respectively, best-score loss, finite loss constraints, the corrected transport budgets, and (3). Thus $H(\Pi)/(1 + \mu^\top z)$ is a lower bound on $V(z)$.

Theorem 2 (Exact representation and convexity)**.** *For every* $z \in [0,1]^D$,

$$v(z) = \max_{\Pi,\mu \text{ satisfying (6)}} \frac{H(\Pi)}{1+\mu^\top z}. \tag{7}$$

The maximum is attained, v *is convex, and* $v(z) = V(z)$ *on binary selections.*

The preceding inequalities explain why each fraction is a valid lower bound. Equality with v requires more: minimizing the Lagrangian of (4) gives the row masses, transport inequalities and one normalization equation. Rescaling those multipliers gives exactly (7). Appendix B derives the objective and every constraint by eliminating one group of variables at a time. For fixed (Π, μ), the fraction is convex because its numerator is nonnegative and its denominator is a positive affine function. The maximum of these convex lower functions is therefore convex.

5 Choosing which binary subset to evaluate

Fix a requested risk accuracy $\varepsilon > 0$. Let $\delta \le \varepsilon/4$ be the allowed inner risk gap, and $\xi \le \varepsilon/4$ the allowed master candidate gap, both nonnegative. Previous evaluations help choose the next binary subset. Every chosen subset then receives a full solve of (4), with all continuous variables reoptimized.

Solve the subset and store a global cut. At $z^s \in \mathcal{Z}_d$, the inner solve returns a finite feasible solution (a^s, u^s, λ^s) and its whole-space test. Its verified risk bound is $R_s = J(a^s, \lambda^s)$. Retaining the constant test ensures $R_s \le B$. The dual solve returns a pair (Π^s, μ^s) satisfying (6). Write its verified fraction as $f_s(z) = h_s/[1 + (\mu^s)^\top z]$, with $0 \le h_s \le H(\Pi^s)$. The certified value $\ell_s \le f_s(z^s)$ must satisfy

$$\ell_s \le V(z^s) \le R_s, \qquad 0 \le R_s - \ell_s \le \delta \le \varepsilon/4.$$

In exact arithmetic $\ell_s = f_s(z^s)$; any numerical downward shift is included in this gap. The fraction's gradient gives the cut

$$g_s = -\frac{h_s \mu^s}{[1 + (\mu^s)^\top z^s]^2}, \qquad L_s(z) = \ell_s + g_s^\top (z - z^s) \le v(z). \tag{8}$$

Convexity puts the tangent below f_s, while (7) puts f_s below v. Store this cut and the actual feasible test together.

Rank a small set of binary proposals. After t full evaluations, the actual test bounds and retained cuts give

$$U_t = \min\{B, R_1, \dots, R_t\}, \qquad a_t(z) = \max\{0, L_1(z), \dots, L_t(z)\}.$$

Here U_t is the global upper bound and $a_t(z)$ is a lower estimate of each subset's optimal risk. For $t = 0$, set $U_0 = B$ and $a_0 = 0$.

Fit a cheap Gaussian-process predictor to the observed pairs (z^s, R_s). A fixed similarity between subsets lets an evaluated subset inform nearby subsets, which receive lower predictive uncertainty when more strongly constrained by these observations. Denote the predicted risk by $m_t(z)$ and its predictive standard deviation by $\sigma_t(z)$. Both are used for ranking, not as guaranteed risk bounds; their construction is given in Appendix E.

Form a small finite pool $\mathcal{C}_t \subseteq \mathcal{Z}_d$ using a few coordinate swaps and random proposals. Keep subsets whose lower estimate allows sufficient improvement, and choose

$$z^{t+1} \in \operatorname*{argmin}_{\substack{z \in \mathcal{C}_t \\ a_t(z) \le U_t - \varepsilon/2}} \big[m_t(z) - \sigma_t(z)\big].$$

The score favors low predicted risk while allowing uncertain subsets to be tried. The screen exceeds the inner error $\delta \le \varepsilon/4$, so a resolved subset cannot be chosen again. The chosen subset is then fully solved: only its actual R_s can update U_t, and only its verified dual result supplies a new cut. The predictor does neither.

Use the global master to certify stopping. The master minimizes the lower model over all allowed subsets:

$$\min_{z\in Z_d,\eta} \eta \quad \text{subject to} \quad \eta \ge 0, \qquad \eta \ge L_s(z) \quad (s = 1, \dots, t). \tag{9}$$

It returns a binary z_{M} and a certified global lower bound. Update LB_t with the maximum of its previous value and this bound; all earlier cuts remain. Stop if $U_t - \mathrm{LB}_t \le \varepsilon$. Otherwise require $a_t(z_{\mathrm{M}}) - \mathrm{LB}_t \le \xi \le \varepsilon/4$ after the update. Then

$$a_t(z_{\mathrm{M}}) \le \mathrm{LB}_t + \xi < U_t - \varepsilon + \xi \le U_t - 3\varepsilon/4,$$

so the master supplies an eligible candidate. Refresh it every three full evaluations and whenever the pool has no eligible subset. If a solve cannot attain its required accuracy within the budget, return the available bounds and test with that status.

Algorithm 1 Full inner solves with prediction-based candidate selection

Require: $\varepsilon > 0$; $0 \le \delta \le \varepsilon/4$; $0 \le \xi \le \varepsilon/4$.

Set $t = 0$, $U_0 = B$, $\mathrm{LB}_0 = 0$; retain the constant test and initialize empty histories and the predictor.
while $U_t - \mathrm{LB}_t > \varepsilon$ **do**
  Form the small binary pool $\mathcal{C}_t$ and keep proposals with $a_t(z) \le U_t - \varepsilon/2$.
  if no proposal is eligible, or three full evaluations have occurred since the last master solve **then**
    Solve (9) and update LB_t with its certified bound.
    If $U_t - \mathrm{LB}_t \le \varepsilon$, return the retained best test; otherwise require $a_t(z_{\mathrm{M}}) - \mathrm{LB}_t \le \xi$ and add z_{M} to the eligible pool.
  end if
  Choose an eligible proposal minimizing $m_t(z) - \sigma_t(z)$.
  Set $s = t + 1$, $z^s = z$; fully solve (4) there, obtaining a feasible test and a verified fraction with $R_s - \ell_s \le \delta$.
  Compute (8) at this evaluated point; store its cut, subset and actual test bound; retain every previous cut.
  Update the predictor with the completed observation (z^s, R_s).
  Set $U_{t+1} = \min\{U_t, R_s\}$, $\mathrm{LB}_{t+1} = \mathrm{LB}_t$, and $t \leftarrow t + 1$.
end while
return The retained best test and subset, with interval $[\mathrm{LB}_t, U_t]$.

The evaluation bound uses the coordinate movement in the data. Let $d_{i\ell,j}$ be entry j of $\mathbf{d}_{i\ell}$ and define

$$M_j = \max_{k=0,1} \frac{\sum_{i\in I_k} q_i \max_\ell d_{i\ell,j}}{\theta_k}, \qquad \rho(z, z') = \sum_j M_j |z_j - z'_j|.$$

Thus ρ measures how selections differ in transport units. Clipping a feasible correction to $0 \le \mu^s \le M$ preserves feasibility; use it before computing the value and gradient in (8). The positive separation scale and number of retained coordinate weights are

$$r_\varepsilon = \frac{\varepsilon/2 - \delta}{B} > 0, \qquad M_{(1)} \ge \cdots \ge M_{(D)}, \qquad s_\varepsilon = \min\left\{ s \in \{0, \dots, D\} : \sum_{j>s} M_{(j)} < r_\varepsilon \right\}.$$

The proof shows that evaluated subsets are at least r_ε apart in ρ; s_ε makes the omitted weight smaller than this distance. Let T_ε count completed full binary evaluations begun before stopping, including the first. Refinements at one selected point count as one evaluation.

Theorem 3 (Global optimality and evaluation bound)**.** *Algorithm 1, with the stated feasibility, correction and accuracy requirements and all generated cuts retained, maintains*

$$\mathrm{LB}_t \le \mathrm{OPT} \le U_t. \tag{10}$$

If the required solves complete without a premature budget exit, it returns a feasible test with robust risk at most $\mathrm{OPT} + \varepsilon$*. The number of completed evaluations satisfies*

$$T_\varepsilon \le \min\left\{|\mathcal{Z}_d|, \sum_{i=0}^{\min(d,s_\varepsilon)} \binom{s_\varepsilon}{i}\right\} \le 2^{s_\varepsilon}.$$

Appendix C proves these statements. The predictor affects which eligible subset is tried, not the validity of the bounds or the count.

6 Loss choices and the scope of the result

The loss enters through Φ in the continuous problem and through Ψ in the dual objective. Examples covered by the same derivation are shown below; boundary values use continuous limits.

Loss	$\Phi(s)$	$\Psi(p_0, p_1)$
Half hinge	$\frac{1}{2}(1+s)_+$	$\min(p_0, p_1)$
Logistic	$\log(1+e^s)$	$(p_0+p_1)\log(p_0+p_1) - p_0\log p_0 - p_1\log p_1$
Exponential	e^s	$2\sqrt{p_0 p_1}$
Squared hinge	$(1+s)_+^2$	$4p_0p_1/(p_0+p_1)$

For example, half-hinge masses $p_0 = 0.2$ and $p_1 = 0.5$ leave loss 0.2: the best common score favors the larger mass but cannot distinguish the smaller mass at the same location. Appendix D derives all four formulas. For half hinge, a score clipped to $[-1, 1]$ gives randomized class-zero probability $(1 + g)/2$, so $V/2$ is the equal-prior worst-case error. Other losses define surrogate objectives. The current implementation uses half hinge and logistic; the other rows illustrate the theoretical scope.

The loss formulas need not be fixed before deriving the algorithm. The transport geometry still matters: the whole-space construction uses the triangle inequality, and the coordinate-price formulation uses the L^1 decomposition. The certified interval concerns the stated robust optimization problem; whether its uncertainty sets cover an unknown population is a separate statistical question.

A Proof of Theorem 1: the finite formulation

Use the assumptions of Section 1, and let $T(z)$ denote the infimum of the finite program on the right of (2). We prove both value inequalities and construct a whole-space test, then establish attainment.

A.1 Finite transports give the lower value bound

Fix binary z and $S = \{j : z_j = 1\}$. Write P_S for the distribution of the selected coordinates under P. The selected distance, projection, and lifting satisfy

$$\begin{aligned} d_z(x, x') &:= \sum_{j=1}^{D} z_j |x_j - x'_j| = \|x_S - x'_S\|_1 \le \|x - x'\|_1, \\ W_1(P_S, \widehat{P}_{k,S}) &\le W_1(P, \widehat{P}_k), \\ \|x_i - (x_{\ell,S}, x_{i,S^c})\|_1 &= d_z(x_i, x_\ell), \\ g(x_{\ell,S}, x_{i,S^c}) &= g(x_\ell). \end{aligned} \tag{A.1}$$

Lifting retains the unselected source coordinates, so it preserves both selected transport cost and test loss. If source projections coincide, split their transported mass according to their weights and retain their source identities. The same construction lifts arbitrary projected targets and preserves finite first moments.

For scores $u = (u_\ell)_{\ell=1}^N$, define $Q_k(u)$ as the largest class-k loss obtained by sending mass to the projected pool. The variable $\Pi_{i\ell}$ sends mass from source $i \in I_k$ to target ℓ; only class-k rows occur in this calculation:

$$\begin{aligned} Q_k(u) = \max_{\Pi \ge 0} \quad & \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ \text{subject to} \quad & \sum_{\ell} \Pi_{i\ell} = q_i \quad (i \in I_k), \\ & \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \le \theta_k. \end{aligned} \tag{A.2}$$

Staying at the source is feasible, and $0 \le \Pi_{i\ell} \le q_i$ makes the feasible set compact. To derive the dual, assign unrestricted row multipliers a_i and a nonnegative budget multiplier λ_k. Because this is a maximization problem, the Lagrangian is

$$\begin{aligned} \mathcal{L}_k &= \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) + \sum_{i \in I_k} a_i \left(q_i - \sum_{\ell} \Pi_{i\ell} \right) \\ &\quad + \lambda_k \left(\theta_k - \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \right) \\ &= \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k + \sum_{i \in I_k, \ell} \Pi_{i\ell} [\Phi(y_i u_\ell) - a_i - \lambda_k d_z(x_i, x_\ell)]. \end{aligned} \tag{A.3}$$

For a feasible transport the row terms vanish and the budget term is nonnegative, so this Lagrangian bounds its loss above. After these constraints enter the Lagrangian, its supremum retains only $\Pi \ge 0$. A positive mass coefficient gives $+\infty$; if every coefficient is nonpositive, $\Pi = 0$ attains the supremum:

$$\sup_{\Pi \ge 0} \mathcal{L}_k = \begin{cases} \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k, & \Phi(y_i u_\ell) \le a_i + \lambda_k d_z(x_i, x_\ell) \quad (i \in I_k,\ \forall \ell), \\ +\infty, & \text{otherwise.} \end{cases} \tag{A.4}$$

Minimize this upper bound over $a_i \in \mathbb{R}$ and $\lambda_k \geq 0$. Finite linear programming strong duality gives

$$Q_k(u) = \min_{\substack{a_i \in \mathbb{R},\ \lambda_k \geq 0 \\ \Phi(y_i u_\ell) \leq a_i + \lambda_k d_z(x_i, x_\ell)\ (i \in I_k,\ \forall \ell)}} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right]. \tag{A.5}$$

The two classes share u, so the last equality gives $T(z) = \inf_u [Q_0(u) + Q_1(u)]$. For any original test, define $u(g) = (g(x_1), \ldots, g(x_N))$. Lift each feasible transport using (A.1): its full-space distribution lies in $\mathcal{U}_k$ and its loss is $\sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i g(x_\ell))$. Thus

$$Q_k(u(g)) \leq \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g). \tag{A.6}$$

Restricting the infimum to induced scores and using this inequality gives

$$\begin{aligned} T(z) &= \inf_u [Q_0(u) + Q_1(u)] \\ &\leq \inf_g [Q_0(u(g)) + Q_1(u(g))] \leq V(z). \end{aligned} \tag{A.7}$$

Separate scores at coincident projected targets only relax the first infimum.

A.2 A whole-space test gives the reverse value bound

Take any feasible (a, u, λ) in (2). To satisfy every source allowance at each x, we seek a test with class-k loss below their minimum:

$$\begin{aligned} E_k(x) &= \min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}, \\ i_k(x) &= \min \arg\min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}. \end{aligned} \tag{A.8}$$

Choose the smallest source index in a tie. At fixed x, abbreviate $i_k = i_k(x)$; c is the distance between the chosen sources and r_k their distances to x:

$$c = d_z(x_{i_0}, x_{i_1}), \qquad r_k = d_z(x_{i_k}, x), \qquad c \leq r_0 + r_1. \tag{A.9}$$

When $c > 0$, blend their scores by

$$\alpha = \frac{\min(c, r_0)}{c}, \qquad g(x) = (1-\alpha) u_{i_0} + \alpha u_{i_1}. \tag{A.10}$$

The triangle inequality ensures that both portions of this distance fit within r_0, r_1:

$$\begin{aligned} 0 \leq \alpha \leq 1, \qquad & \alpha c = \min(c, r_0) \leq r_0, \\ & (1-\alpha)c = c - \min(c, r_0) = \max(c - r_0, 0) \leq r_1. \end{aligned} \tag{A.11}$$

In particular, the last inequality uses $c - r_0 \leq r_1$ and $r_1 \geq 0$.

Substitute source–target pairs (i_0, i_0), (i_0, i_1), (i_1, i_0), and (i_1, i_1) in (2). Their source signs are -1 and $+1$, and the own-source distances are zero:

$$\begin{aligned} \Phi(-u_{i_0}) &\leq a_{i_0}, & \Phi(-u_{i_1}) &\leq a_{i_0} + \lambda_0 c, \\ \Phi(u_{i_0}) &\leq a_{i_1} + \lambda_1 c, & \Phi(u_{i_1}) &\leq a_{i_1}. \end{aligned} \tag{A.12}$$

Convexity, these four constraints, and (A.11) give the two pointwise loss bounds:

$$\begin{aligned} \Phi(-g(x)) &\leq (1-\alpha)\Phi(-u_{i_0}) + \alpha \Phi(-u_{i_1}) \\ &\leq (1-\alpha) a_{i_0} + \alpha(a_{i_0} + \lambda_0 c) = a_{i_0} + \lambda_0 \alpha c \\ &\leq a_{i_0} + \lambda_0 r_0 = E_0(x), \\ \Phi(g(x)) &\leq (1-\alpha)\Phi(u_{i_0}) + \alpha \Phi(u_{i_1}) \\ &\leq (1-\alpha)(a_{i_1} + \lambda_1 c) + \alpha a_{i_1} = a_{i_1} + \lambda_1 (1-\alpha) c \\ &\leq a_{i_1} + \lambda_1 r_1 = E_1(x). \end{aligned} \tag{A.13}$$

If $c = 0$, set $g(x) = u_{i_0}$. The first and third endpoint inequalities give instead

$$\Phi(-g(x)) \le a_{i_0} \le E_0(x), \qquad \Phi(g(x)) \le a_{i_1} + \lambda_1 c = a_{i_1} \le E_1(x). \tag{A.14}$$

Finite comparisons of continuous functions make $i_k(x)$ Borel measurable, and hence the piecewise g is Borel measurable. It uses only x_S and lies between $\min_\ell u_\ell$ and $\max_\ell u_\ell$. It extends the loss bounds; interpolation of every prescribed u_ℓ is unnecessary.

Now integrate these bounds. Fix $P \in \mathcal{U}_k$ and choose a coupling of cost at most θ_k, retaining source indices and splitting coincident atoms according to their weights. Let γ_k be its measure on $I_k \times \mathbb{R}^D$, with target marginal P. Its source rows and cost satisfy

$$\gamma_k(\{i\} \times \mathbb{R}^D) = q_i, \qquad \int d_z(x_i, x)\, d\gamma_k(i,x) \le \int \|x_i - x\|_1\, d\gamma_k(i,x) \le \theta_k. \tag{A.15}$$

For each source $i \in I_k$, the constructed score satisfies $\Phi((2k-1)g(x)) \le E_k(x) \le a_i + \lambda_k d_z(x_i, x)$. Using both marginals gives

$$\begin{aligned}
\mathbb{E}_P \Phi((2k-1)g) &= \int \Phi((2k-1)g(x))\, d\gamma_k(i,x) \\
&\le \int a_i\, d\gamma_k(i,x) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&= \sum_{i \in I_k} a_i \gamma_k(\{i\} \times \mathbb{R}^D) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&= \sum_{i \in I_k} q_i a_i + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&\le \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k.
\end{aligned} \tag{A.16}$$

The bound is independent of P, so taking the classwise suprema and adding yields

$$\sum_{k=0}^{1} \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g) \le \sum_{k=0}^{1} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right] = J(a, \lambda).$$

Every finite feasible point therefore yields an original test with risk at most J. Taking the infimum gives $V(z) \le T(z)$; (A.7) gives the other direction. This proves the value equality and feasible-test claim in Theorem 1.

A.3 Attainment of the primal minima

The following argument proves attainment for Theorem 1 and for the risk program used in Theorem 2. Nonconstancy and monotonicity give $s_1 < s_2$ with $\Phi(s_2) > \Phi(s_1)$. Set $b = [\Phi(s_2) - \Phi(s_1)]/(s_2 - s_1) > 0$; convexity gives, for $t > s_2$,

$$\frac{\Phi(t) - \Phi(s_2)}{t - s_2} \ge b \quad \Longrightarrow \quad \Phi(t) \ge \Phi(s_2) + b(t - s_2) \longrightarrow \infty. \tag{A.17}$$

The constant point $u = 0$, $a_i = \Phi(0)$, $\lambda = 0$ is feasible in (2) with $J = B$. Adding $w_0 = w_1 = 0$ and $\eta = B$ makes it feasible in (4) for every $z \in [0,1]^D$. Thus the objective sublevels $J \le B$ and $J \le \eta \le B$ are nonempty. Own-source constraints give $a_i \ge \Phi(y_i u_i) \ge 0$; positive weights and radii then give

$$\begin{aligned}
0 \le q_i a_i \le q^\top a \le J \le B \quad &\Longrightarrow\ 0 \le a_i \le B/q_i, \\
0 \le \theta_k \lambda_k \le \theta^\top \lambda \le J \le B \quad &\Longrightarrow\ 0 \le \lambda_k \le B/\theta_k, \\
0 \le w_{kj} \le \lambda_k \le B/\theta_k, \quad &0 \le J \le \eta \le B \quad \text{in (4).}
\end{aligned} \tag{A.18}$$

Choose a fixed source $i_k \in I_k$ for each class. The finite pool distances and (A.18) bound both score signs by the finite constant

$$C = \max_{k,\ell} \left\{ B/q_{i_k} + (B/\theta_k)\|\mathbf{d}_{i_k\ell}\|_1 \right\},$$

$$\Phi((2k-1)u_\ell) \le B/q_{i_k} + (B/\theta_k)\|\mathbf{d}_{i_k\ell}\|_1 \le C.$$

Choose $R > 0$ with $\Phi(R) > C$, possible by (A.17). Monotonicity gives

$$u_\ell \ge R \implies \Phi(u_\ell) > C, \qquad u_\ell \le -R \implies \Phi(-u_\ell) > C; \qquad |u_\ell| < R.$$

All variables on these sublevels are bounded. The finite convex loss is continuous, so the sublevels are closed and hence compact. Both continuous objectives attain their minima there. Since a constant test with value B is already feasible, restricting to these sublevels preserves both global minima.

B Proof of Theorem 2: the dual representation

B.1 Multipliers and the Lagrangian

Fix $z \in [0,1]^D$ in (4). The variables η, a, u are free, whereas λ, w_0, w_1 remain nonnegative. Write each remaining constraint with left side at most zero and attach the multiplier shown below:

Constraint	Nonnegative multiplier
$\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} \le 0$	$\bar{\Pi}_{i\ell}$
$J - \eta \le 0$	τ
$w_{kj} - \lambda_k \le 0$	$\bar{\beta}_{kj}$
$\theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j \le 0$	$\bar{\mu}_j$

Here $\bar{\Pi} \in \mathbb{R}_+^{N\times N}$, $\tau \in \mathbb{R}_+$, and $\bar{\beta}_k, \bar{\mu} \in \mathbb{R}_+^D$. The bars distinguish these multipliers from the row-normalized matrix and correction vector in the main text.

Adding multiplier times constraint to the objective gives

$$\begin{aligned} \mathcal{L} = \eta + \tau(J-\eta) &+ \sum_{i,\ell} \bar{\Pi}_{i\ell}[\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i}] \\ &+ \sum_k \bar{\beta}_k^\top (w_k - \lambda_k \mathbf{1}) + \bar{\mu}^\top(\theta_0 w_0 + \theta_1 w_1 - \eta z). \end{aligned} \tag{B.1}$$

At a feasible point of (4), every added constraint term is nonpositive. Therefore, with the multipliers fixed,

$$\inf_{\substack{\eta\in\mathbb{R},\ a,u\in\mathbb{R}^N \\ \lambda,w_0,w_1\ge 0}} \mathcal{L} \le \mathcal{L}(\eta, a, u, \lambda, w_0, w_1) \le \eta.$$

The infimum is a lower bound on $v(z)$. The dual problem maximizes this lower bound over the multipliers; equality will be verified below. All four constraint groups are now in $\mathcal{L}$; they are not imposed again when taking its infimum. In particular, the nonnegativity of a and η implied by the original constraints does not restrict their domains in this infimum.

To collect the score and price terms, define

$$\bar{p}_{k\ell} = \sum_{i\in I_k} \bar{\Pi}_{i\ell}, \qquad A_k(\bar{\Pi}) = \sum_{i\in I_k,\ell} \bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}. \tag{B.2}$$

The first quantity is the mass of class k at target ℓ; the second is its vector of coordinate movements. The regrouping uses

$$\begin{aligned}\sum_{i,\ell}\bar{\Pi}_{i\ell}a_i &= \sum_i a_i \sum_\ell \bar{\Pi}_{i\ell},\\ \sum_{i,\ell}\bar{\Pi}_{i\ell}\Phi(y_i u_\ell) &= \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)],\\ \sum_{i,\ell}\bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}^\top w_{c_i} &= \sum_k w_k^\top A_k(\bar{\Pi}),\\ \tau J &= \sum_i \tau q_i a_i + \sum_k \tau\theta_k\lambda_k.\end{aligned} \tag{B.3}$$

Thus each primal variable group has its own term:

$$\begin{aligned}\mathcal{L} = \eta(1-\tau-\bar{\mu}^\top z) &+ \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell})\\ &+ \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)]\\ &+ \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top\bar{\beta}_k) + \sum_k w_k^\top[\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})].\end{aligned} \tag{B.4}$$

B.2 Minimizing over the primal variables

Keep all multipliers fixed. The dual value is the infimum of (B.4) over $\eta \in \mathbb{R}$, $a, u \in \mathbb{R}^N$ and $\lambda, w_0, w_1 \geq 0$. Since these domains are independent, minimize each term separately.

For a linear coefficient b, the two rules are

$$\inf_{t\in\mathbb{R}} bt = \begin{cases}0, & b = 0,\\ -\infty, & b \neq 0,\end{cases} \qquad \inf_{t\geq 0} bt = \begin{cases}0, & b \geq 0,\\ -\infty, & b < 0.\end{cases}$$

For the first rule, $t = -nb$ gives $bt = -nb^2 \to -\infty$ when $b \neq 0$. For the second, $t = n$ gives $bt \to -\infty$ when $b < 0$; otherwise the minimum is zero at $t = 0$.

The risk variable and source allowances. Applying the free-variable rule gives

$$\begin{aligned}\inf_{\eta\in\mathbb{R}} \eta(1-\tau-\bar{\mu}^\top z) > -\infty \iff 1-\tau-\bar{\mu}^\top z = 0,\\ \inf_{a\in\mathbb{R}^N}\sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell}) > -\infty \iff \tau q_i - \sum_\ell \bar{\Pi}_{i\ell} = 0 \quad (\forall i).\end{aligned} \tag{B.5}$$

Both terms then contribute zero. The second set of equalities is $\bar{\Pi}\mathbf{1} = \tau q$.

The scores. One score u_ℓ serves the two masses at target ℓ. By the definition of Ψ in (5),

$$\begin{aligned}\inf_{u\in\mathbb{R}^N}\sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)] &= \sum_\ell \inf_{s\in\mathbb{R}}[\bar{p}_{0\ell}\Phi(-s) + \bar{p}_{1\ell}\Phi(s)]\\ &= \sum_\ell \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) = H(\bar{\Pi}).\end{aligned} \tag{B.6}$$

To justify the first equality even at boundary masses, fix $\varepsilon > 0$ and choose each u_ℓ so that

$$\Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) \le \bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell) \le \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) + \varepsilon/N.$$

Summing puts the loss between $H(\bar{\Pi})$ and $H(\bar{\Pi}) + \varepsilon$; let $\varepsilon \downarrow 0$. The infima are finite because

$$0 \le \Psi(p_0, p_1) \le \Phi(0)(p_0 + p_1), \qquad 0 \le H(\bar{\Pi}) \le \Phi(0) \sum_{i,\ell} \bar{\Pi}_{i\ell} = \tau B.$$

The last equality uses the row condition and $\sum_i q_i = 2$.

The class and coordinate prices. Applying the nonnegative-variable rule gives

$$\begin{aligned} \inf_{\lambda \ge 0} \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top \bar{\beta}_k) > -\infty \iff \mathbf{1}^\top \bar{\beta}_k \le \tau\theta_k \quad (\forall k), \\ \inf_{w_0, w_1 \ge 0} \sum_k w_k^\top [\bar{\beta}_k + \theta_k \bar{\mu} - A_k(\bar{\Pi})] > -\infty \iff A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k \bar{\mu} \quad (\forall k). \end{aligned} \tag{B.7}$$

Each term contributes zero when its condition holds. Therefore the minimized Lagrangian is $H(\bar{\Pi})$ exactly under (B.5) and (B.7), and is $-\infty$ otherwise. Maximizing this finite value over nonnegative multipliers is the dual problem.

It remains to remove $\bar{\beta}_k$. Coordinatewise, its conditions imply

$$\begin{aligned} \bar{\beta}_{kj} \ge 0, \quad A_{kj}(\bar{\Pi}) \le \bar{\beta}_{kj} + \theta_k \bar{\mu}_j \iff \bar{\beta}_{kj} \ge \max(0, A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j), \\ \mathbf{1}^\top \bar{\beta}_k \le \tau\theta_k \implies \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \le \tau\theta_k. \end{aligned} \tag{B.8}$$

Conversely, if the last inequality holds, choose

$$\bar{\beta}_k = (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+.$$

This vector is nonnegative, has sum at most $\tau\theta_k$, and satisfies $A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k \bar{\mu}$. Thus both directions are exact, and the dual is

$$\begin{aligned} \max_{\tau, \bar{\Pi}, \bar{\mu} \ge 0} \quad & H(\bar{\Pi}) \\ \text{subject to} \quad & \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = 1, \\ & \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \le \tau\theta_k, \quad k = 0, 1. \end{aligned} \tag{B.9}$$

We next verify equality with the primal optimum and attainment, before dividing by τ.

B.3 Equality of primal and dual values

Primal attainment was established in Appendix A.3.

The restriction to objective values at most B was used only to prove existence; it is not added to (4). For strong duality in that program, first identify the prices already fixed by a zero coordinate:

$$z_j = 0 \implies 0 \le \theta_0 w_{0j} + \theta_1 w_{1j} \le \eta z_j = 0 \implies w_{0j} = w_{1j} = 0.$$

This uses $\theta_k > 0$ and $w_{kj} \ge 0$. Delete these fixed variables and the resulting tautological coordinate inequalities. On the remaining coordinates choose

$$u = 0, \quad a_i = \Phi(0) + 1, \quad \lambda_k = 1, \quad w_k = \delta z, \quad 0 < \delta < 1, \quad \eta = J + 1 + \delta(\theta_0 + \theta_1). \tag{B.10}$$

Here $J = 2[\Phi(0) + 1] + \theta_0 + \theta_1$. Every relevant residual is strictly negative:

$$\begin{aligned} \Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} &= -1 - \delta \mathbf{d}_{i\ell}^\top z < 0, \\ w_{kj} - \lambda_k &= \delta z_j - 1 < 0, \\ J - \eta &= -1 - \delta(\theta_0 + \theta_1) < 0, \\ \theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j &= -(J+1) z_j < 0 \quad (z_j > 0). \end{aligned} \tag{B.11}$$

Also $w_{kj} > 0$ on those coordinates and $\lambda_k > 0$. Finite-dimensional convex strong duality applies to this reduced problem: the dual maximum is attained and equals $v(z)$.

To restore an omitted coordinate, use

$$z_j = 0: \quad \bar{\beta}_{kj} = 0, \qquad \bar{\mu}_j = \max_k A_{kj}(\bar{\Pi})/\theta_k.$$

Then $\bar{\beta}_{kj} + \theta_k \bar{\mu}_j \geq A_{kj}(\bar{\Pi})$ for both classes, no allocation budget increases, and $\bar{\mu}_j z_j = 0$ leaves normalization unchanged. Thus the full dual (B.9) also has an attained maximum equal to $v(z)$.

B.4 Removing the normalization

We first justify division by τ. If $\tau = 0$, nonnegativity and the row condition give

$$\bar{\Pi}\mathbf{1} = 0 \implies \bar{\Pi} = 0 \implies H(\bar{\Pi}) = \sum_\ell \inf_s 0 = 0.$$

For a positive scale, the mass-scaling identity follows directly from the infimum:

$$\begin{aligned} \Psi(\tau p_0, \tau p_1) &= \inf_s \tau[p_0 \Phi(-s) + p_1 \Phi(s)] = \tau \Psi(p_0, p_1), \\ H(\tau \Pi) &= \sum_\ell \Psi(\tau p_{0\ell}, \tau p_{1\ell}) = \tau H(\Pi), \qquad \tau > 0. \end{aligned} \tag{B.12}$$

There is, however, a feasible point with strictly positive objective. Choose a target ℓ_*, and let e_{ℓ_*} be the vector with entry one at that target and zero elsewhere. Sending every source to this target means $\Pi = q e_{\ell_*}^\top$. Its received masses are one from each class, so

$$H(\Pi) = \Psi(1,1), \qquad \Phi(-s) + \Phi(s) \geq 2\Phi(0), \qquad \Psi(1,1) = 2\Phi(0) = B > 0.$$

The last equality follows by taking $s = 0$. Set

$$\mu_j = \max_k \frac{A_{kj}(\Pi)}{\theta_k}, \qquad \tau = \frac{1}{1 + \mu^\top z}, \qquad \bar{\Pi} = \tau \Pi, \qquad \bar{\mu} = \tau \mu. \tag{B.13}$$

These choices satisfy every normalized constraint:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z &= \tau(1 + \mu^\top z) = 1, \\ A_k(\Pi) - \theta_k \mu \leq 0 \implies (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ &= 0, \\ H(\bar{\Pi}) = \tau H(\Pi) &= \tau B > 0. \end{aligned}$$

Thus an attained optimal normalized solution cannot have $\tau = 0$.

Now take any normalized feasible point with $\tau > 0$ and define $\Pi = \bar{\Pi}/\tau$, $\mu = \bar{\mu}/\tau$. The row and movement constraints transform as follows:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q &\iff \Pi\mathbf{1} = q, \\ A_k(\bar{\Pi}) &= \tau A_k(\Pi), \\ (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ &= \tau (A_k(\Pi) - \theta_k \mu)_+, \\ \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \leq \tau \theta_k &\iff \sum_j (A_{kj}(\Pi) - \theta_k \mu_j)_+ \leq \theta_k. \end{aligned} \tag{B.14}$$

The third line uses $(\tau t)_+ = \tau t_+$ for $\tau > 0$. Normalization then gives the denominator and the objective:

$$\begin{aligned} \tau + \bar{\mu}^\top z = 1 \iff \tau(1 + \mu^\top z) = 1 \iff \tau = (1 + \mu^\top z)^{-1}, \\ H(\bar{\Pi}) = \tau H(\Pi) = \frac{H(\Pi)}{1 + \mu^\top z}. \end{aligned} \tag{B.15}$$

The denominator is positive since $\mu, z \geq 0$. Conversely, start with any pair satisfying (6), choose τ as in (B.15), and multiply both objects by τ. Equation (B.14) verifies the row and movement conditions, and $\tau(1 + \mu^\top z) = 1$ verifies normalization. The objectives agree in both directions. This proves (7) and its attained maximum.

B.5 Equality on binary selections

For an original finite feasible point, put $\eta = J$ and $w_k = \lambda_k z$. Then

$$\begin{aligned} 0 \leq w_k \leq \lambda_k \mathbf{1}, \\ \theta_0 w_{0j} + \theta_1 w_{1j} = z_j \theta^\top \lambda \leq z_j J = z_j \eta, \\ a_i + \mathbf{d}_{i\ell}^\top w_{c_i} = a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z. \end{aligned} \tag{B.16}$$

The middle inequality is (3). All constraints of (4) therefore hold, giving $v(z) \leq V(z)$.

Conversely, for a risk-model feasible point, $\theta_k > 0$ and nonnegative prices give

$$\begin{aligned} z_j = 0 \implies \theta_0 w_{0j} + \theta_1 w_{1j} \leq 0 \implies w_{0j} = w_{1j} = 0, \\ z_j = 1 \implies w_{kj} \leq \lambda_k, \\ \implies \quad w_k \leq \lambda_k z. \end{aligned} \tag{B.17}$$

Since $\mathbf{d}_{i\ell} \geq 0$ coordinatewise,

$$\Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i} \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z.$$

Thus (a, u, λ) is feasible in (2), with $V(z) \leq J \leq \eta$. Taking the minimum over risk-model feasible points gives $V(z) \leq v(z)$. Combining both directions yields $v(z) = V(z)$.

B.6 Convexity and completion of Theorem 2

Fix one feasible pair (Π, μ) in (6). Its remaining loss $h = H(\Pi) \geq 0$ is constant when z changes. For $z \in [0,1]^D$, define the scalar correction $r(z) = 1 + \mu^\top z > 0$ and its lower function $f(z) = h/r(z)$. For two coordinates $j, m \in \{1, \ldots, D\}$, the chain rule gives

$$\begin{aligned} \frac{\partial r}{\partial z_j} = \mu_j, \qquad \frac{\partial f}{\partial z_j} = h[-r(z)^{-2}]\mu_j = -\frac{h\mu_j}{r(z)^2}, \\ \frac{\partial^2 f}{\partial z_m \partial z_j} = -h\mu_j[-2r(z)^{-3}]\mu_m = \frac{2h\mu_j\mu_m}{r(z)^3}, \\ \nabla^2 f(z) = \frac{2h\mu\mu^\top}{r(z)^3}. \end{aligned} \tag{B.18}$$

For any direction $\xi \in \mathbb{R}^D$,

$$\xi^\top \nabla^2 f(z) \xi = \frac{2h}{r(z)^3}(\mu^\top \xi)^2 \geq 0.$$

Thus f is convex. Taking the maximum over the fixed feasible pairs in (7) proves that v is convex.

Together with the attained ratio representation and binary equality above, this completes the proof of Theorem 2.

C Proof of Theorem 3: full inner evaluations and global convergence

At every selected binary subset, the full inner evaluation produces a feasible test and a dual certificate. The proof uses the test and tangent obtained in that same solve.

C.1 Data bounds for the certificates

Put

$$D_{kj} = \sum_{i \in I_k} q_i \max_{\ell} d_{i\ell,j}, \qquad M_j = \max_{k=0,1} D_{kj}/\theta_k, \qquad \rho(z,z') = \sum_j M_j |z_j - z'_j|.$$

The row condition $\Pi \mathbf{1} = q$ gives

$$0 \le A_{kj}(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} d_{i\ell,j} \le \sum_{i \in I_k} \Big(\sum_{\ell} \Pi_{i\ell} \Big) \max_{\ell} d_{i\ell,j} = D_{kj} \le \theta_k M_j.$$

If $\mu_j > M_j$, replacing it by M_j leaves the positive part in (6) equal to zero in that coordinate for both classes. It therefore preserves feasibility and raises the lower fraction. Thus the representation in Theorem 2 may be restricted to $0 \le \mu \le M$.

Choosing score zero in (5) bounds every certificate numerator:

$$0 \le H(\Pi) \le \Phi(0) \sum_{\ell} (p_{0\ell} + p_{1\ell}) = 2\Phi(0) = B.$$

C.2 Cuts from full evaluations and valid bounds

Full evaluation s at $z^s \in \mathcal{Z}_d$ produces a finite feasible test with objective $R_s = J(a^s, \lambda^s)$ and a feasible dual pair (Π^s, μ^s). After any certificate repair and clipping, use one common numerator $0 \le h_s \le H(\Pi^s)$ to define

$$f_s(z) = \frac{h_s}{1 + (\mu^s)^\top z}, \qquad \ell_s = f_s(z^s) - e_s, \qquad e_s \ge 0, \qquad \ell_s \le V(z^s) \le R_s, \qquad R_s - \ell_s \le \delta.$$

The nonnegative margin e_s permits a deliberate downward shift of the cut; $e_s = 0$ in ideal arithmetic. Its contribution is included in δ. Here R_s is the objective of an actually feasible test obtained at z^s, not a prediction at that point. The finite-formulation theorem gives its whole-space risk guarantee. The derivative and tangent at this subset are

$$\begin{aligned} g_s &= -\frac{h_s \mu^s}{[1 + (\mu^s)^\top z^s]^2}, \qquad -BM_j \le g_{s,j} \le 0, \\ L_s(z) &= \ell_s + g_s^\top (z - z^s) \\ &= f_s(z^s) + g_s^\top (z - z^s) - e_s \le f_s(z) \le v(z). \end{aligned}$$

The first inequality follows from convexity of the fraction and $e_s \ge 0$. The downward shift leaves its bounded gradient unchanged. Only L_s at its evaluated point z^s is retained. With all these cuts retained, set

$$a_t(z) = \max\{0, L_s(z) : s \le t\} \le V(z).$$

The predictor in Appendix E assigns each proposal a mean $m_t(z)$ and a model standard deviation $\sigma_t(z)$. They determine the evaluation order, by minimizing $m_t(z) - \sigma_t(z)$ among proposals satisfying the cut filter. The actual incumbent is updated only by full evaluations:

$$U_t = \min\{B, R_s : s \le t\}.$$

Consequently OPT $\le U_t$, with a corresponding genuinely feasible test. A certified lower bound on $\min_{z \in \mathcal{Z}_d} a_t(z)$ is a lower bound on OPT. Taking the maximum of zero and all such bounds defines LB_t and proves (10).

C.3 A global fallback always supplies an eligible point

Candidates are accepted only if

$$a_t(z) \le U_t - \varepsilon/2. \tag{C.1}$$

The finite proposal pool and its ranking may otherwise be arbitrary. If it contains no eligible point, solve the current master (9) and update the running lower bound by taking its maximum with the certified bound. Previous lower bounds remain valid because cuts are only added. The verified binary candidate z_M and updated running bound must satisfy

$$\mathrm{LB}_t \le \min_{z \in \mathcal{Z}_d} a_t(z), \qquad a_t(z_M) - \mathrm{LB}_t \le \xi.$$

Stop if $U_t - \mathrm{LB}_t \le \varepsilon$. Otherwise,

$$a_t(z_M) \le \mathrm{LB}_t + \xi < U_t - \varepsilon + \xi \le U_t - 3\varepsilon/4 < U_t - \varepsilon/2.$$

Thus the fallback candidate passes (C.1). Periodic master solves may certify the result earlier, but are not needed for this argument. Validity and termination do not depend on how eligible proposals are ranked.

C.4 Separation, evaluation count and termination

Consider an earlier completed evaluation s and a later accepted subset z^t. Let U_{before} be the incumbent just before selecting z^t. The earlier actual objective remains in the incumbent history, so $U_{\text{before}} \le R_s$. Since its own cut is still retained,

$$\ell_s \ge R_s - \delta \ge U_{\text{before}} - \delta, \qquad L_s(z^t) \le a_{t-1}(z^t) \le U_{\text{before}} - \varepsilon/2.$$

Subtracting these inequalities and using the bounded nonpositive gradient,

$$\begin{aligned} \varepsilon/2 - \delta \le \ell_s - L_s(z^t) &= -g_s^\top (z^t - z^s) \\ &\le B \sum_j M_j (z_j^t - z_j^s)_+ \le B\rho(z^t, z^s). \end{aligned}$$

Thus all completed selections are distinct and their pairwise distances are at least $r_\varepsilon = (\varepsilon/2 - \delta)/B > 0$.

Sort $M_{(1)} \ge \cdots \ge M_{(D)}$ and choose

$$s_\varepsilon = \min\left\{ s \in \{0, \dots, D\} : \sum_{j>s} M_{(j)} < r_\varepsilon \right\}.$$

If two selections agree on these s_ε coordinates, then

$$\rho(z, z') \le \sum_{j > s_\varepsilon} M_{(j)} < r_\varepsilon,$$

contradicting the separation just proved. Every completed evaluation therefore has a different pattern on the retained coordinates. Because a feasible selection uses at most d of them, counting these patterns gives

$$T_\varepsilon \le \min\left\{ |\mathcal{Z}_d|, \sum_{i=0}^{\min(d, s_\varepsilon)} \binom{s_\varepsilon}{i} \right\} \le 2^{s_\varepsilon}.$$

The strict inequality in the omitted-coordinate sum is necessary because the separation allows equality. This count includes the first full evaluation; refinements needed to achieve its required gap are not extra evaluations.

If the algorithm continued after exhausting these patterns, the global fallback would supply another eligible subset. Its completed evaluation would require another distinct pattern, a contradiction. Hence it reaches $U_t - \mathrm{LB}_t \le \varepsilon$. For the returned feasible test, whose actual robust risk is denoted by R,

$$R \le U_t \le \mathrm{LB}_t + \varepsilon \le \mathrm{OPT} + \varepsilon.$$

The bound concerns completed accuracy-controlled evaluations, not inner iterations or wall time. If an inner solve or a master solve cannot meet its required accuracy, return the available feasible test and valid bounds without claiming this termination guarantee. In particular, a master incumbent objective is not a certified lower bound. Certificate errors and candidate-comparison tolerances must be included in the stated accuracy budgets. In particular, an affine downward shift is included in the evaluated gap and need not change the gradient.

D Loss formulas for Section 6

Write $F(s) = p_0\Phi(-s) + p_1\Phi(s)$, so $\Psi(p_0, p_1) = \inf_s F(s)$. First take $p_0, p_1 > 0$; the axes are treated below. When the minimum is attained, s^* denotes a minimizing scalar score.

For half hinge, $\Phi(s) = (1+s)_+/2$. If $s > 1$, replacing it by 1 keeps $\Phi(-s) = 0$ and reduces $\Phi(s)$. If $s < -1$, replacing it by -1 keeps $\Phi(s) = 0$ and reduces $\Phi(-s)$. It is therefore sufficient to minimize on $[-1, 1]$:

$$\begin{aligned} F(s) &= \frac{p_0(1-s) + p_1(1+s)}{2} = \frac{p_0 + p_1}{2} + \frac{p_1 - p_0}{2}s, \\ \Psi(p_0, p_1) &= \begin{cases} F(-1) = p_0, & p_1 \ge p_0, \\ F(1) = p_1, & p_1 < p_0, \end{cases} = \min(p_0, p_1). \end{aligned} \tag{D.1}$$

The choice of endpoint follows from the sign of the linear coefficient; when $p_0 = p_1$, every score in the interval is optimal.

For logistic, $\Phi(s) = \log(1 + e^s)$. Differentiate the two terms separately:

$$\begin{aligned} \frac{d}{ds}\left[p_0 \log(1 + e^{-s})\right] &= -\frac{p_0 e^{-s}}{1 + e^{-s}} = -\frac{p_0}{1 + e^s}, \\ \frac{d}{ds}\left[p_1 \log(1 + e^{s})\right] &= \frac{p_1 e^{s}}{1 + e^{s}}, \\ F'(s) = \frac{-p_0 + p_1 e^s}{1 + e^s} = 0 \quad &\Leftrightarrow \quad -p_0 + p_1 e^{s^*} = 0 \\ &\Leftrightarrow \quad e^{s^*} = p_0/p_1, \quad s^* = \log(p_0/p_1). \end{aligned}$$

The derivative changes from negative to positive at s^*, so this is the minimum. Substituting both logarithms gives

$$\begin{aligned} 1 + e^{-s^*} &= (p_0 + p_1)/p_0, \qquad 1 + e^{s^*} = (p_0 + p_1)/p_1, \\ \Psi(p_0, p_1) &= p_0 \log\frac{p_0 + p_1}{p_0} + p_1 \log\frac{p_0 + p_1}{p_1} \\ &= p_0[\log(p_0 + p_1) - \log p_0] + p_1[\log(p_0 + p_1) - \log p_1] \\ &= (p_0 + p_1)\log(p_0 + p_1) - p_0 \log p_0 - p_1 \log p_1. \end{aligned} \tag{D.2}$$

For exponential, $\Phi(s) = e^s$. A square proves the lower bound and identifies equality:

$$\begin{aligned} F(s) - 2\sqrt{p_0 p_1} &= \left(\sqrt{p_0}e^{-s/2} - \sqrt{p_1}e^{s/2}\right)^2 \ge 0, \\ \sqrt{p_0}e^{-s^*/2} = \sqrt{p_1}e^{s^*/2} \quad &\Leftrightarrow \quad e^{s^*} = \sqrt{p_0/p_1}, \quad s^* = \tfrac{1}{2}\log(p_0/p_1), \\ \Psi(p_0, p_1) &= F(s^*) = 2\sqrt{p_0 p_1}. \end{aligned} \tag{D.3}$$

For squared hinge, $\Phi(s) = (1+s)_+^2$. The same clipping argument restricts the score to $[-1,1]$. Expanding and completing the square gives

$$\begin{aligned} F(s) &= p_0(1-s)^2 + p_1(1+s)^2 \\ &= (p_0+p_1)s^2 + 2(p_1-p_0)s + (p_0+p_1) \\ &= (p_0+p_1)\left(s - \frac{p_0-p_1}{p_0+p_1}\right)^2 + (p_0+p_1) - \frac{(p_0-p_1)^2}{p_0+p_1}, \end{aligned} \tag{D.4}$$

$$(p_0+p_1) - \frac{(p_0-p_1)^2}{p_0+p_1} = \frac{(p_0+p_1)^2-(p_0-p_1)^2}{p_0+p_1} = \frac{4p_0p_1}{p_0+p_1},$$

$$s^* = \frac{p_0-p_1}{p_0+p_1} \in [-1,1], \qquad \Psi(p_0,p_1) = F(s^*) = \frac{4p_0p_1}{p_0+p_1}.$$

The nonnegative square vanishes at this feasible score, which proves the minimum.

On the axes, nonnegativity and the indicated score limits give

$$0 \le \Psi(p_0,0) \le p_0\Phi(-s) \longrightarrow 0 \quad (s \to +\infty),$$
$$0 \le \Psi(0,p_1) \le p_1\Phi(s) \longrightarrow 0 \quad (s \to -\infty).$$

Consequently,

$$\Psi(p_0,0) = \lim_{s\to+\infty} p_0\Phi(-s) = 0, \qquad \Psi(0,p_1) = \lim_{s\to-\infty} p_1\Phi(s) = 0, \qquad \Psi(0,0) = 0.$$

For half hinge and squared hinge, the first zero is already attained at $s \ge 1$, and the second at $s \le -1$. For logistic and exponential with a positive remaining mass, it is only an infimum as the score tends in the stated direction. The closed formulas use $0 \log 0 = 0$ and assign the squared-hinge value zero when both masses are zero.

E Risk predictions for the selection rule

The predictor uses only the pairs (z^s, R_s) from completed inner evaluations. It assigns greater similarity to subsets differing in fewer coordinates. The following model determines their evaluation order; the bounds and termination in Theorem 3 do not require it to describe the true risk function correctly.

E.1 Predicting risk and uncertainty

Normalize observed risks by B, use prior mean $1/2$, and set

$$r_t = (R_1/B, \dots, R_t/B)^\top,$$
$$k(z,z') = \tfrac{1}{4}\exp\left(-\frac{\|z-z'\|_1}{\max(1,d)}\right), \qquad \nu = \max\{10^{-8}, (\delta/B)^2\}.$$

The parameter ν regularizes the prediction calculation. Let I_t be the identity matrix and $\mathbf{1}_t$ the vector of ones. Define

$$K_t = (k(z^s,z^r))_{s,r=1}^t, \qquad k_t(z) = (k(z^1,z), \dots, k(z^t,z))^\top, \qquad A_t = K_t + \nu I_t.$$

For a fixed proposal z, let Y model the normalized observations and let F model its normalized risk. Use the auxiliary joint Gaussian model

$$\begin{pmatrix} Y \\ F \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \frac{1}{2}\mathbf{1}_t \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} A_t & k_t(z) \\ k_t(z)^\top & \frac{1}{4} \end{pmatrix}\right).$$

The binary kernel is positive definite. Each coordinate contributes the matrix $\left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$, where $r = \exp[-1/\max(1, d)]$. Its eigenvalues are $1 \pm r > 0$. The full binary kernel is their tensor product, scaled by $1/4$. For distinct evaluated subsets, K_t is a principal submatrix of that matrix. The joint covariance is positive definite: the kernel contribution is nonnegative, while νI_t adds a strictly positive contribution whenever the observation component is nonzero; a vector with only its last component nonzero has positive variance $1/4$. Consequently A_t is invertible and its Schur complement $1/4 - k_t(z)^\top A_t^{-1} k_t(z)$ is positive. Put

$$E = F - \tfrac{1}{2} - k_t(z)^\top A_t^{-1}(Y - \tfrac{1}{2}\mathbf{1}_t).$$

Its covariance with the observed vector and its variance are

$$\begin{aligned} \mathrm{Cov}(E, Y) &= k_t(z)^\top - k_t(z)^\top A_t^{-1} A_t = 0, \\ \mathrm{Var}(E) &= \tfrac{1}{4} - k_t(z)^\top A_t^{-1} k_t(z). \end{aligned}$$

Thus E and Y are independent under this Gaussian model. Conditioning on the actual observed vector $Y = r_t$ gives the risk prediction and its model standard deviation:

$$\begin{aligned} m_t(z) &= B\left[\tfrac{1}{2} + k_t(z)^\top A_t^{-1}(r_t - \tfrac{1}{2}\mathbf{1}_t)\right], \\ \sigma_t(z) &= B\sqrt{\tfrac{1}{4} - k_t(z)^\top A_t^{-1} k_t(z)}. \end{aligned} \tag{E.1}$$

For $t = 0$, these are $m_0 = B/2$ and $\sigma_0 = B/2$. The standard deviation measures uncertainty in this prediction model; it is not a certified confidence bound for the true risk. Only a new full inner solve supplies a new feasible test.

E.2 Updating the predictor with little overhead

Maintain the lower-triangular Cholesky factor C_t of A_t. Appending one completed evaluation requires only

$$\begin{aligned} A_t &= C_t C_t^\top, \qquad C_t v_* = k_t(z^{t+1}), \\ c &= \sqrt{\tfrac{1}{4} + \nu - v_*^\top v_*}, \qquad C_{t+1} = \begin{pmatrix} C_t & 0 \\ v_*^\top & c \end{pmatrix}. \end{aligned}$$

Indeed,

$$\begin{aligned} C_{t+1} C_{t+1}^\top &= \begin{pmatrix} A_t & C_t v_* \\ v_*^\top C_t^\top & v_*^\top v_* + c^2 \end{pmatrix} \\ &= \begin{pmatrix} A_t & k_t(z^{t+1}) \\ k_t(z^{t+1})^\top & \frac{1}{4} + \nu \end{pmatrix} = A_{t+1}. \end{aligned}$$

The new diagonal is positive because the augmented matrix is positive definite. Prediction uses triangular solves rather than matrix inversion:

$$C_t p = r_t - \tfrac{1}{2}\mathbf{1}_t, \qquad C_t^\top \alpha_t = p, \qquad C_t v = k_t(z),$$

so that

$$m_t(z) = B[\tfrac{1}{2} + k_t(z)^\top \alpha_t], \qquad \sigma_t(z) = B\sqrt{\max\{0, \tfrac{1}{4} - v^\top v\}}.$$

The factor update and each bounded batch of predictions cost $O(t^2)$ arithmetic operations, in addition to $O(Dt)$ work for each new similarity vector. Kernel parameters are fixed; there is no hyperparameter optimization between inner solves. The maximum with zero in the variance formula only removes numerical roundoff.